Reagents

Cell preparation and crosslinking
- Formaldehyde (Thermo Scientific, 28906)
- PBS, pH7.4
- Glycine, 2.5M stock
- Bovine Serum Albumin (BSA) 10%

Nuclei permeabilization and tagmentation
- AMBER permeabilization buffer (see Reagent preparation)
- AMBER barcoded Tn5 (see Reagent preparation)
- Tagmentation Buffer (see Reagent preparation)
- EDTA, 40.4 mM (Thermo Scientific, 15575020)

Lysis and collection of chromatin complexes
- AMBER lysis buffer (see Reagent preparation)
- AMBER wash buffer, 0 M NaCl and 0.15 M NaCl (see Reagent preparation)
- IGEPAL CA-630, 10% (Sigma, I8896)
- Streptavidin C1 beads (Thermo Scientific, 65001)
- Magnetic separation rack (NEB, S1515S)

Gap-filling and elution
- KAPA HiFi HotStart ReadyMix Kit (Roche, 50-196-5299)
- Nuclease-free water

AMBER hydrogel and reactions
- AMBER Spatial Barcode (molecular beacon)
- 4-arm PEG acrylate, 10K MW
- HS-PEG-SH, 3.4K MW
- Phi29 DNA polymerase and Phi29 reaction buffer
- DMSO
- BSA
- dNTP mix
- HaeIII
- Nb.BsrDI
- CutSmart Buffer (NEB)
- T7 DNA Ligase (NEB)
- StickTogether DNA Ligase Buffer (NEB)

Gel denaturation, DNA extraction, PCR and sequencing
- AMBER gel denaturation buffer, high pH (see Reagent preparation)
- AMBER gel neutralization buffer (see Reagent preparation)
- ThermoLabile Proteinase K (NEB, P8111S)
- KAPA HiFi HotStart ReadyMix Kit (Roche, 50-196-5299)
- Illumina-indexed forward and reverse primers
- SPRI beads

Oligonucleotides (all 5’-3’)

AMBER Seed Oligo Primer:
TAGGAACATCAAACGACAGCCA
AMBER Seed Oligo Template:
/5Phos/TCGTTTGATGTTCCTAGGCCGCATGCNNVNNVNNHNNVNNVCTGTCTCTTATACACATCTG
ACGCTGCCGACGAGGCCGAGCATAGAGGGATTGGCTG
Tn5 transposome AMBER Adapter A:
GCATGCGGCATTGCAGATGTGTATAAGAGACAG
Tn5 transposome Adapter B:
GTCTCGTGGGCTCGGAGATG/iBiodT/GTATAAGAGACAG
pMENTs: CTGTCTCTTATACACATCT

Experimental Reagents Preparation

Please prepare the following reagents fresh, alongside their storage conditions:

AMBER Wash Buffer (4°C)

No Salt: Tris-HCl pH 7.5 (10mM), EDTA (1mM), NaCl (0mM), Tween-20 (0.1%)
Low Salt: Tris-HCl pH 7.5 (10mM), EDTA (1mM), NaCl (150mM), Tween-20 (0.1%)

AMBER Lysis Buffer (4°C)

Tris-HCl pH8 (10mM), EDTA (1mM), NaCl (200mM), Triton X-100 (1%), SDS (0.75%), SDOC (0.75%), EDTA-free Protease Inhibitor Cocktail (1X)

AMBER Gel Denaturation Buffer (25°C)

KOH (457.5mM), EDTA (100mM), dTT (42.5mM)

AMBER Neutralization Buffer (25°C)

Tris HCl, pH 7.5 (600mM), HCl (305mM)

OMNI Permeabilization Buffer - (ON ICE)

ATAC-RSB with 0.10% IGEPAL, 0.10% Tween20, 0.01% Digitonin, RNAseOUT, SUPERaseIN, and a 50X PI Cocktail.

Tagmentation Buffer (2X) - (4°C)

Tris-Ac pH7.8 (66mM), KAc (133mM), MgAc (22mM), DMF (32%).

AMBER Seed Oligos

1. Combine the circularisation reaction reagents on ice (20µL total): 2µL AMBER Seed Oligo Template (5µM; final 0.5µM) + 1µL CircLigase ssDNA ligase (100 U/µL) + 1µL ATP (1mM; final 50µM) + 1µL MnCl2 (50mM; final 2.5mM) + 2µL CircLigase reaction buffer (10x; final 1x) + 13µL H2O.
1b. Please include a CircLigase ssDNA control reaction alongside as a positive control.
2. Incubate for 3h at 60°C, then 10min at 80°C to inactivate the ligase, then cool to 4 °C.
3. Verify circularisation by gel electrophoresis: run 2µL of the circularised product next to 2µL of the uncircularised template and the control pair (ladder, test, test + CircLigase, control, control + CircLigase).
4. Add 0.5µL Exonuclease I (10U, 20U/µL stock) to the 20µL reaction (20.5µL total). Incubate for 1h at 37°C, then 10min at 80°C, then cool to 4°C.
5. Add 20.5µL AMBER Seed Oligo Primer (0.5µM) and 20.5µL TE buffer pH8 (final conc 61.5µL)
6. Stand at room temperature for 2hr. Store at -20 °C <12 months.

Transposon Tn5 adapter loading

1. Mix the oligo as: 25µL AMBER Adaptor A (100µM), 25µL pMENTs (100µM).
2. Heat the mixture for 5 mins at 95°C and slowly cool down to 20°C at the speed of 0.1°C/s.
3. Mix 5µL of annealed transposome DNA with 30µL of unloaded Tn5 (0.5 mg/mL), briefly vortex and quickly spun down.
4. Incubate the mixture at room temperature for 30mins then at 4°C for 10 minutes.
5. Store at -20°C for <6 months.

Experimental Procedure

Sample Processing

Cell preparation and crosslinking

1. Prepare single-cell suspension of 1 million HeLa cells, or primary tissue mouse cortex cells.
2. Resuspend cells in 0.2% formaldehyde diluted in PBS, pH7.4, and incubate for 10 minutes at room temperature with end-to-end rotation.
3. Quench with 2.5 M glycine to a final concentration of 0.2 M. Incubate for 5 minutes at room temperature, then 5 minutes on ice.
4. Centrifuge at 1,000 g for 10 minutes at 4 °C and remove the supernatant.
5. Wash the crosslinked cells once with chilled 0.5% BSA in PBS, and centrifuge at 1,000 g for 10 minutes at 4 °C.

Nuclei permeabilization

1. Resuspend cells in 200µL of chilled permeabilization buffer per 500,000 cells and incubate on ice for 5 minutes.
2. Quantify intact nuclei by counting a small aliquot.
3. Centrifuge nuclei at 1,000 g for 10 minutes at 4 °C and discard the supernatant.

AMBER

Tn5 tagmentation

1. Aliquot approximately 250,000 permeabilized nuclei per reaction tube and resuspend in 50µL Tagmentation Buffer.
2. Add 2.3µL of AMBER barcoded Tn5 mix and mix gently.
3. Incubate at 37 °C for 35 minutes with agitation at 550 rpm on a ThermoMixer.
4. Stop the reaction by adding 5µL of 40.4 mM EDTA.
5. Pool tagmented nuclei, centrifuge at 1,000 g for 10 minutes at 4 °C, and remove the supernatant.

Nuclei lysis and Chromatin isolation

1. Vigorously resuspend 400,000 tagmented nuclei (about 50 pipette strokes) in 96µL AMBER lysis buffer to release chromatin complexes.
2. Incubate at 37 °C for 1.5 hours with shaking at 800 rpm (ThermoMixer).
3. Validate lysis by counting a small aliquot under a microscope against a no-lysis control.

Collection of regulatory chromatin complexes

1. Add 64µL of AMBER wash buffer (Low Salt) and 20µL of 10% IGEPAL CA-630 to each lysate tube and mix.
2. Wash 10µL of Streptavidin C1 beads three times with AMBER wash buffer (Low Salt), add them to the lysate, and incubate for 1.5-2 hours at room temperature with end-to-end rotation.
3. Spin the tube down gently, place it on a magnetic separation rack, and carefully discard the supernatant.
4. Wash the beads twice, gently, with 200µL AMBER wash buffer (Low Salt), fully resuspending off the rack.

Important: Handle the complexes gently in all downstream steps to preserve large chromatin complexes. Avoid high-speed centrifugation, as insoluble large complexes could pellet and be lost.

Gap-filling

1. Resuspend the beads in 100µL of 1x KAPA HiFi HotStart ReadyMix
2. Incubate at 72°C for 10 minutes at 1,050 rpm on a ThermoMixer. This fills the gaps and heat-displaces the Tn5-derived ME oligo.
3. Wash the beads once with AMBER wash buffer (Low Salt), then three times with AMBER wash buffer (No Salt).

Important: Fully resuspend the beads off the rack to remove magnesium ions from the gap-filling buffer. Leftover magnesium strongly inhibits elution in the next step and results in inefficient recovery of chromatin complexes.

Elution

1. Resuspend the washed beads in 10µL nuclease-free water.
2. Elute by heating slowly to 80°C (ramp rate 0.5°C/s) and holding for 45 seconds.
3. Place the beads on a magnet and transfer the supernatant containing the chromatin complexes to a new tube.
4. Determine the concentration with a spectrophotometer or Qubit.
5. Set aside a 1µL aliquot for decrosslinking. Use Qubit quantification and PCR amplification as proxies for the quality of the chromatin complexes.

Note: We typically recover 2,500-3,000 ng of total decrosslinked, cleaned regulatory DNA per 1,000,000 initial input cells as per spectrophotometer.

AMBER hydrogel preparation

1. Prepare the hydrogel master mix on ice.
2. For a 10µL reaction, combine the following to the final amounts listed:
   - chromatin fragments, about 400 ng (by spectrophotometer)
   - 0.83µL Phi29 DNA polymerase, 0.83 U
   - 1µL Phi29 reaction buffer, 1x
   - 0.42µL DMSO, 4.2%
   - 0.42µL BSA, 4.2 mg/mL
   - 0.58µL AMBER Seed Oligo with Primer, 19 nM
   - 1.67µL 4-arm PEG acrylate, 10K MW, 1.67µL
   - 1.67µL HS-PEG-SH, 3.4K MW, 1.67µL
3. Incubate at 30 °C for 30 minutes with a heated lid (100°C) to polymerise the gel.

Important: Please prepare hydrogel mixes on ice or in a 4°C cold room to prevent immature polymerization of PEG gel components. Accordingly, add the gel components last, and mix well upon adding to master mix.

AMBER hydrogel enzyme reactions

1. Gently overlay dNTP mix (2.25uL of 5.5nM dNTP, 4.4X CutSmart Buffer) on the polymerised gel and incubate 2 hours at 4 °C.
2. Perform rolling circle amplification at 37°C for 2 hours, then incubate 65°C for 20 minutes.
3. Overlay the dual-enzyme cutting mix (4.5µL; 15 U HaeIII and 15 U Nb.BsrDI in 3.3x CutSmart Buffer). Incubate at 37 °C for 10 hours, then 80 °C for 20 minutes.
4. Overlay the ligation mix (5.5µL; 4,500 U T7 DNA Ligase in 1.45x StickTogether DNA Ligase Buffer). Incubate at 20 °C for 20 hours.
5. Place the gel on ice, remove any supernatant, and wash briefly (30s) with 100uL of chilled PBS to remove residual reagents.

Gel denaturation

1. To 40% of the gel volume (about 8µL), add and mix chilled high-pH AMBER gel denaturation buffer to degrade the gel crosslinks by alkaline hydrolysis.
2. Incubate for 2 hours at 4°C, pipetting intermittently (twice, spaced) to ensure thorough denaturation.
3. Neutralize with 12µL of AMBER gel neutralization buffer at room temperature and spin down lightly.

DNA extraction, PCR and sequencing

1. Wash 10µL of Streptavidin C1 beads as before and resuspend in 80µL AMBER wash buffer (both Low Salt). Add them to the liquefied gel and incubate end-to-end for 1 hour at room temperature to recollect the complexes.
2. Isolate the beads on a magnet, discard the supernatant, and wash completely twice with AMBER wash buffer (Low Salt).
3. Resuspend the dried beads in 20µL PBS with 10uL ThermoLabile Proteinase K. Incubate at 37 °C, 1,400 rpm on a ThermoMixer for 8-16 hours.
4. Incubate 55°C for 15 minutes at 1,000 rpm.
5. Isolate the beads on a magnet and use the supernatant directly for PCR.
6. Set up PCR with separate Illumina-indexed forward and reverse primers for each AMBER gel, in 1x KAPA HiFi HotStart ReadyMix
7. PCR cycle as follows:
   - 72 °C for 5 min
   - 98 °C for 30 s
   - 98 °C for 20 s, 63 °C for 20 s, 72 °C for 1 min (repeated)
   - 72 °C for 2.5 min
   - hold at 12 °C
8. Run an initial 9 cycles. Take a 2µL aliquot for qPCR to determine the number of cycles needed to reach 20 nM.
9. Clean the final library with 0.8x (44µL) SPRI beads and elute in 15µL nuclease-free water.

Finish